Clinical Natural Language Technology for Health Care:

Past, Present, and Future Approaches

Defining the Topic

Clinical Natural Language Processing (NLP) refers to the set of technologies that convert unstructured clinical text like physician notes, discharge summaries, claims narratives, and scanned records into structured, machine-readable data. This field has evolved through three overlapping eras. Early systems relied on rule-based pattern matching: dictionaries of medical abbreviations and regular expressions that extracted specific fields, such as medication dosages, from text. The next era introduced statistical and deep-learning models, including named entity recognition (NER) and optical character recognition (OCR), which allowed systems to generalize beyond hand-written rules and to digitize scanned or handwritten documents. The current era is defined by large language models (LLMs) and large multimodal models (LMMs), which can read free text and even document images directly, reason across context, and produce structured output without task-specific training. Computer vision now complements this pipeline by interpreting document layouts, scanned forms, and embedded tables that pure text models cannot process alone.

Relevant Trends

The clinical NLP and healthcare LLM markets are both in a period of rapid, sustained growth. Industry analysts project the healthcare and life sciences NLP market to grow from roughly $5.18 billion in 2025 to $16.01 billion by 2030, a compound annual growth rate of approximately 25%, driven primarily by the volume of unstructured data trapped in electronic health records and the need to extract value from it (MarketsandMarkets, 2026). Several concrete trends stand out:

- Ambient and generative documentation: Vendors such as Microsoft (Nuance DAX Copilot) and Google Cloud (MedLLM) are embedding LLMs directly into clinical documentation and EHR workflows, automating notes and reducing clinician administrative burden (MarketsandMarkets, 2026).
- Multimodal integration: Newer platforms increasingly combine clinical text with imaging and lab data in a single model, improving predictive analytics and decision support (Healthcare LLM Platforms Market Forecast, 2026).
- Administrative automation: NLP is increasingly applied to billing, coding, and prior authorization, directly reducing operational costs for payers and providers (Healthcare LLM Platforms Market Forecast, 2026).

Opportunities

- Faster, deeper claims review: NLP and LLMs can extract diagnosis, procedure, and treatment detail directly from clinical documentation attached to a claim, allowing payment integrity reviewers to validate medical necessity and coding accuracy without manually reading full charts.
- Scalable policy-to-data matching: Structured extraction allows automatic comparison between what a clinical note documents and what a claim bills for, surfacing discrepancies that point to coding errors or overpayment risk at a scale manual review cannot match.
- Reduced provider abrasion: More accurate, automated pre-payment review reduces false-positive flags on legitimate claims, supporting Cotiviti's existing emphasis on minimizing friction with providers during claim edits.

Threats and Risks

- Hallucination and factual errors: LLMs can generate plausible-sounding but incorrect clinical claims. One adversarial benchmark found hallucination rates as high as 65.9% under default prompting conditions, falling to 44.2% only after active mitigation strategies were applied (PMC, 2026). In a payment-integrity context, an undetected hallucination could trigger an incorrect claim denial or approval.
- Regulatory and compliance exposure: Clinical text contains protected health information, so any NLP pipeline must maintain strict HIPAA compliance; this is consistently cited as a primary constraint on deployment speed across the industry (Decoding Market Trends in NLP in Healthcare, 2026).
- Audit and explainability requirements: Because medical hallucinations often use coherent, domain-specific language, they can be difficult for non-expert reviewers to catch, which raises the bar for human-in-the-loop verification and explainable model output (Medical Hallucination in Foundation Models, 2026).

Strategic Recommendation for Cotiviti

Cotiviti's core business already depends on validating claims against medical records, payment policies, and contract terms at large scale. Clinical NLP is therefore not a peripheral innovation for Cotiviti, it is a direct extension of the existing payment-accuracy workflow. Two specific, achievable investments follow from this analysis:

1. A hybrid rule-based and LLM extraction layer for clinical record review. Rather than relying on LLMs alone, Cotiviti should pair fast, deterministic, auditable rule-based extraction for well-defined fields like dosage, diagnosis codes, and dates with LLM-based extraction reserved for ambiguous or novel phrasing the rules miss. This hybrid

design, which mirrors the accompanying proof of concept, limits hallucination exposure while preserving the flexibility LLMs offer over purely manual review.

2. A human-in-the-loop confidence-scoring pipeline. Given the documented hallucination risk in clinical LLM applications, any production deployment should score extraction confidence and automatically route low-confidence or high-dollar-impact cases to human reviewers, rather than fully automating claim decisions. This approach captures the majority of efficiency gains from automation while containing the patient-safety and compliance risk inherent to current-generation LLMs.

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